

Project Proposal: Membership Inference Attacks on Differentially Private Image Classifiers

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Problem Statement. Machine learning models may leak information about **individual training samples**. A well-known privacy risk is the membership inference attack (MIA), where an attacker tries to determine whether a specific sample was part of the training set. This is especially relevant in privacy-sensitive applications, since even **black-box access to model predictions** may reveal information about the underlying data. In this project, we study how effectively differentially private training can mitigate such leakage. More specifically, we ask: *How do privacy budget, attack type, and model architecture affect membership leakage in differentially private image classification?*

Approach. We will train image classifiers under two settings: (1) standard non-private training using SGD/Adam and (2) differentially private training using DP-SGD with a systematic scan over different privacy budgets ϵ . We will compare at least two image classifiers, such as a small CNN and a deeper CNN. We will evaluate both black-box membership inference attacks based on model outputs (e.g. confidence- and loss-based attacks) and at least one white-box attack based on internal model information such as gradient norms. **We will compare the resulting models using test accuracy, train-test accuracy gap, attack AUC, TPR at low FPR, and training runtime.** Our goal is to study the privacy-utility trade-off and examine whether DP-SGD remains effective against stronger attacks and across different model architectures. We will also analyze how overfitting and training dynamics relate to attack success.

Datasets, Algorithms, and Libraries.

- **Datasets:** CIFAR-10 subset; Fashion-MNIST/MNIST for sanity check
- **Algorithms:** image classification, DP-SGD, black-box and white-box MIA
- **Libraries:** PyTorch, Opacus, NumPy, pandas, scikit-learn, Matplotlib